

Mapterhorn Terrain and Imagery

CNG Zürich 2026

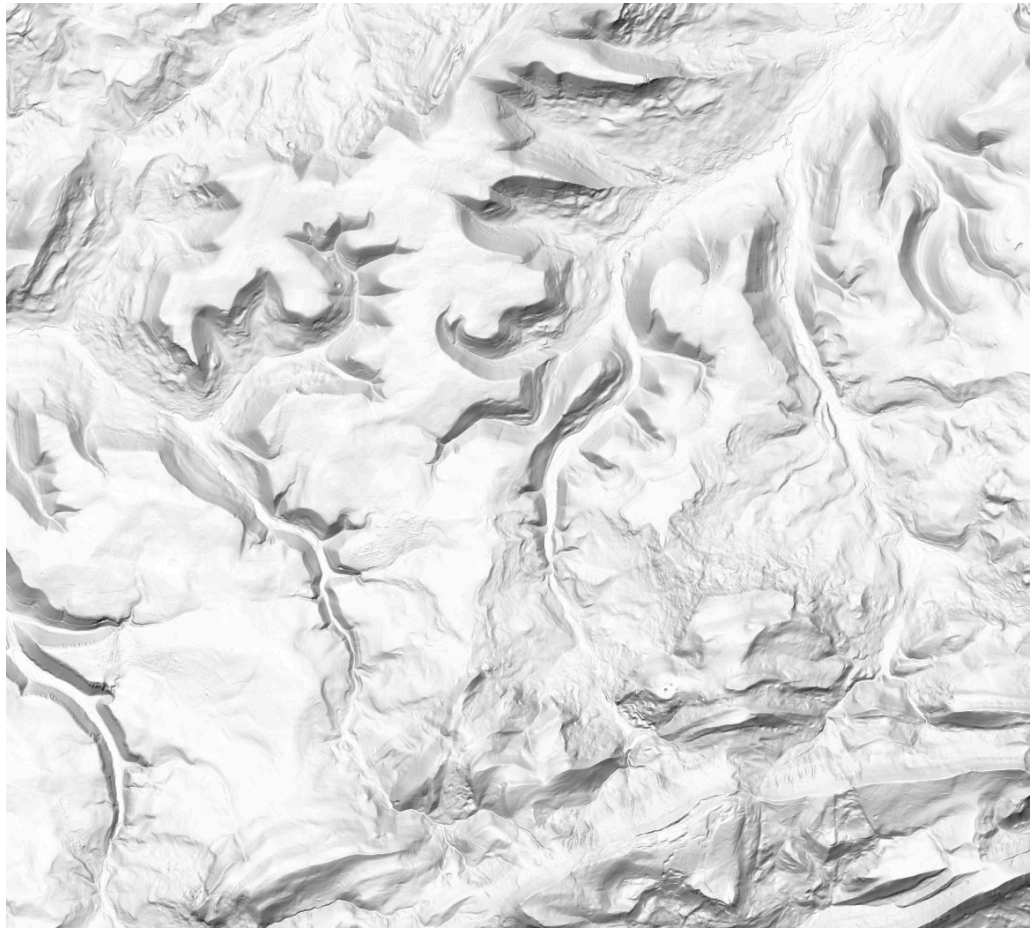
Zürich · Oct 29th, 2026

Oliver Wipfli



About Mapterhorn

- Aggregation of open terrain data
- Global low-resolution coverage, local high-resolution lidar
- For hillshading, contour lines, routing, u.s.w.
- Soon also aerial imagery



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About Me

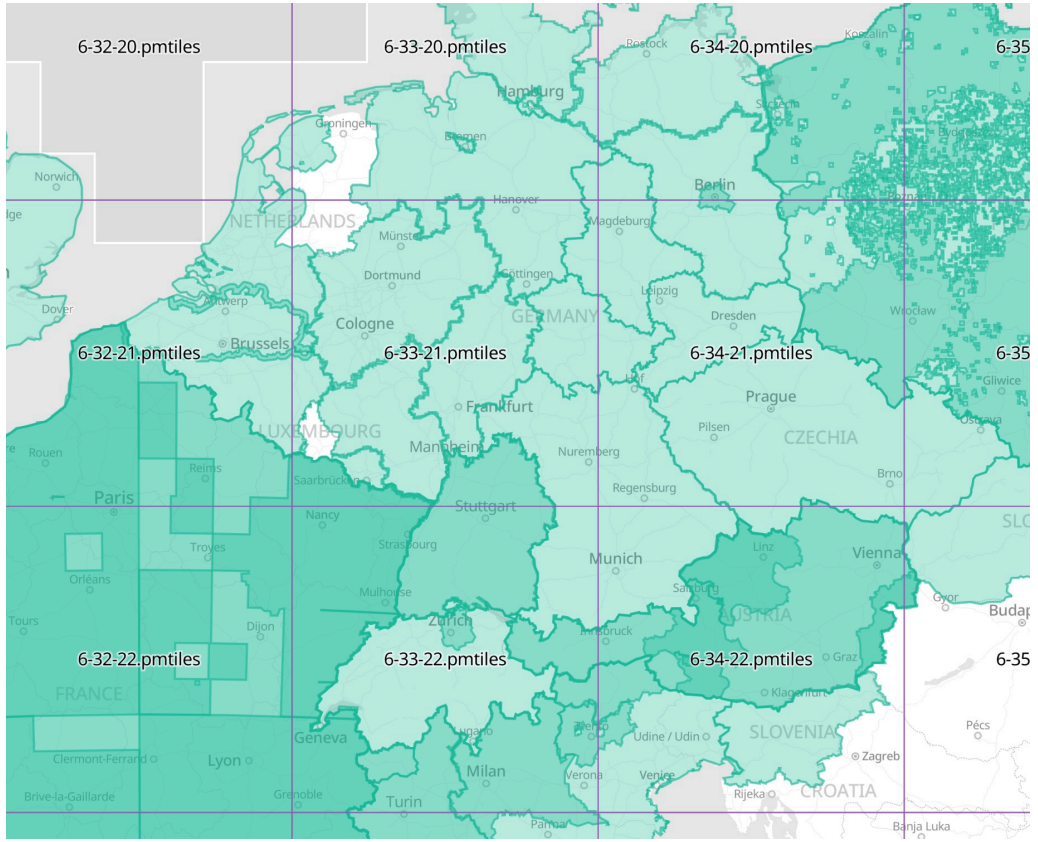
- Oliver Wipfli
- Based in Zürich, Switzerland
- Consultant in Maps and Location Technology
- Email: oliver.wipfli@leichteralsluft.ch
- GitHub: https://github.com/wipfli
- LinkedIn: oliver-wipfli-562258210



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Data Providers

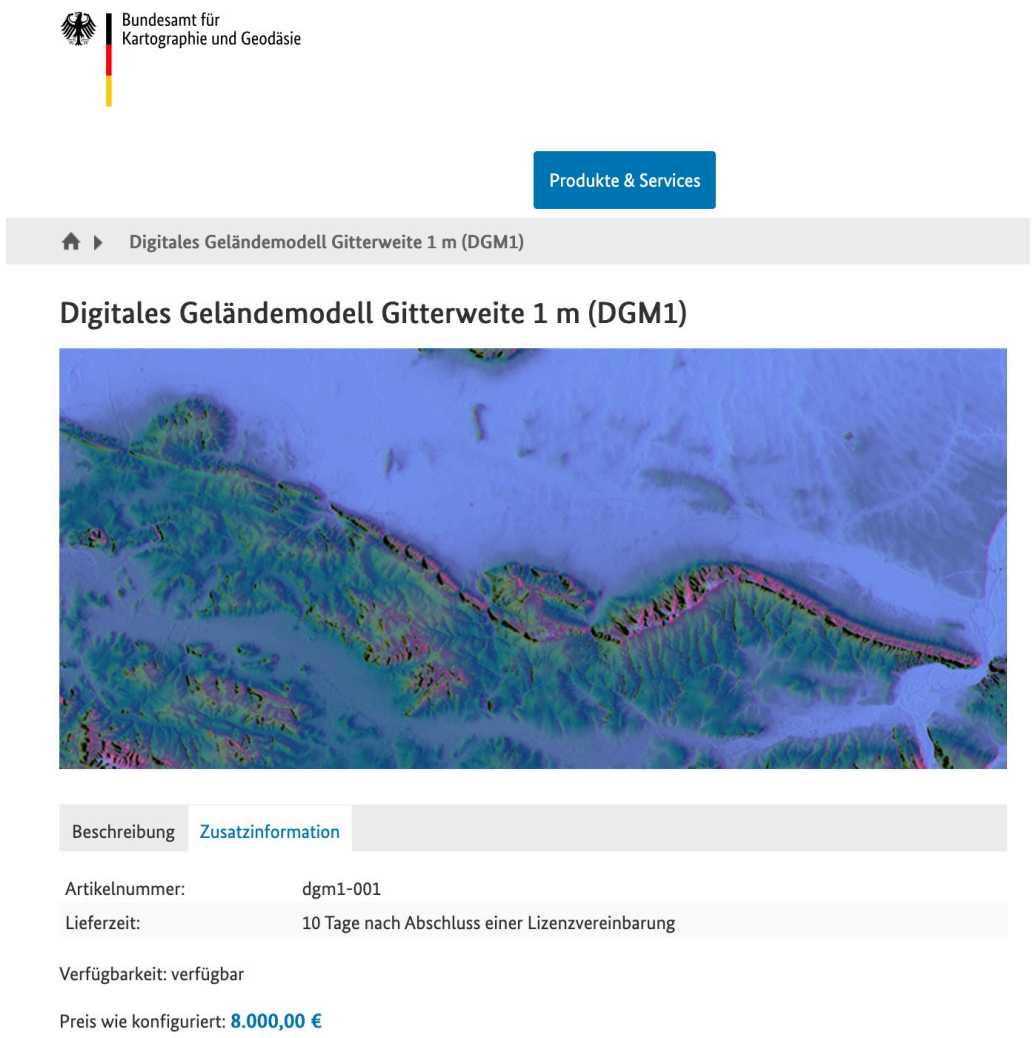
- Global: Copernicus 30 m resolution
- Local: High-resolution Lidar
- Start with Switzerland…



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Germany

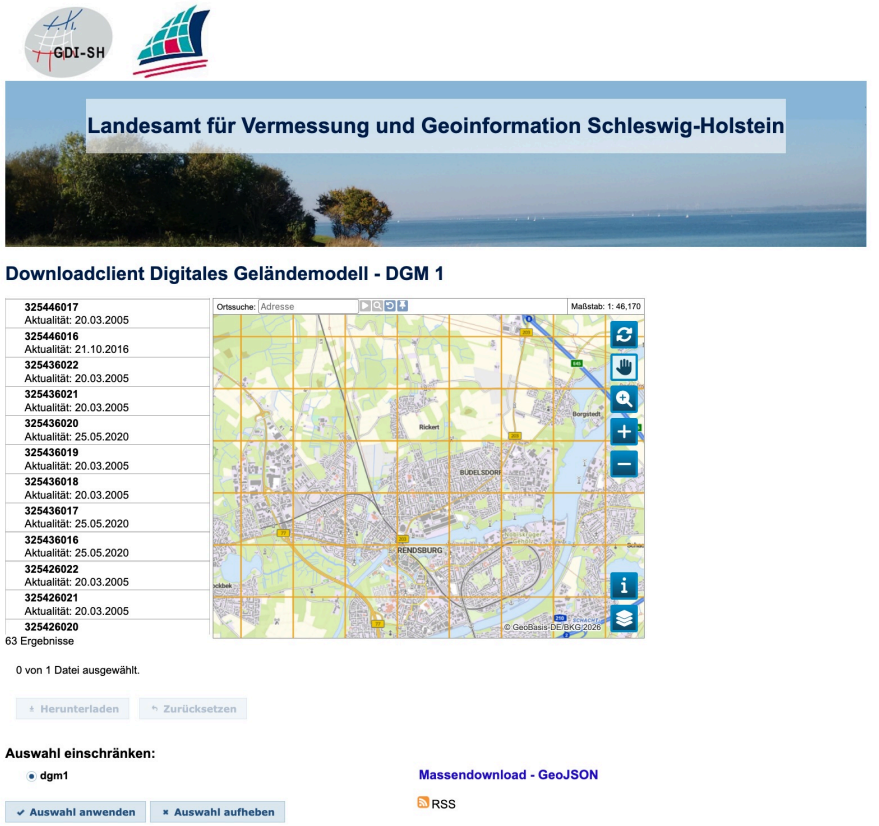
- DGM1, 1 meter per pixel, entire country
- But: EUR 8000



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16 Bundesländer

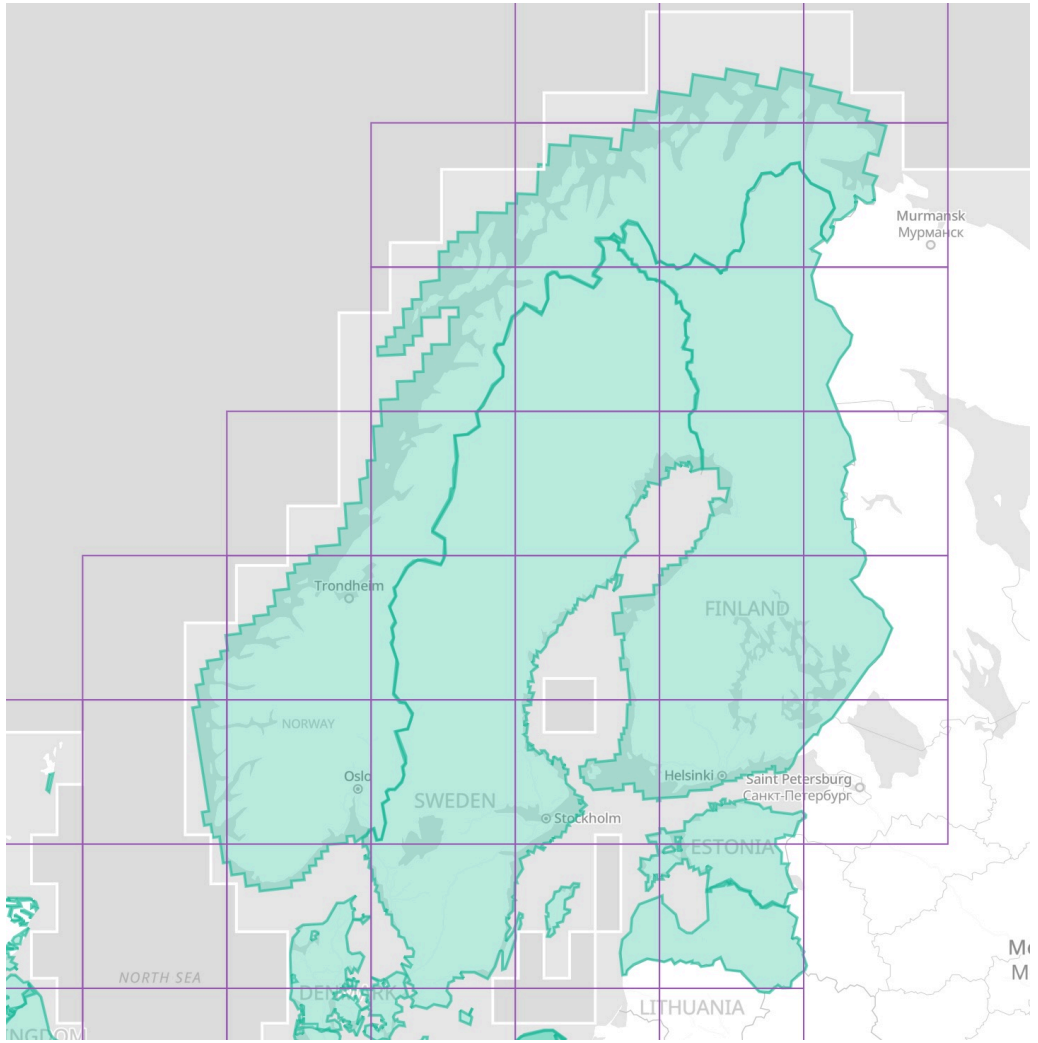
- One by one…
- Best: file lists (STAC, meta4, atom)
- Worst: click to download



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Sweden

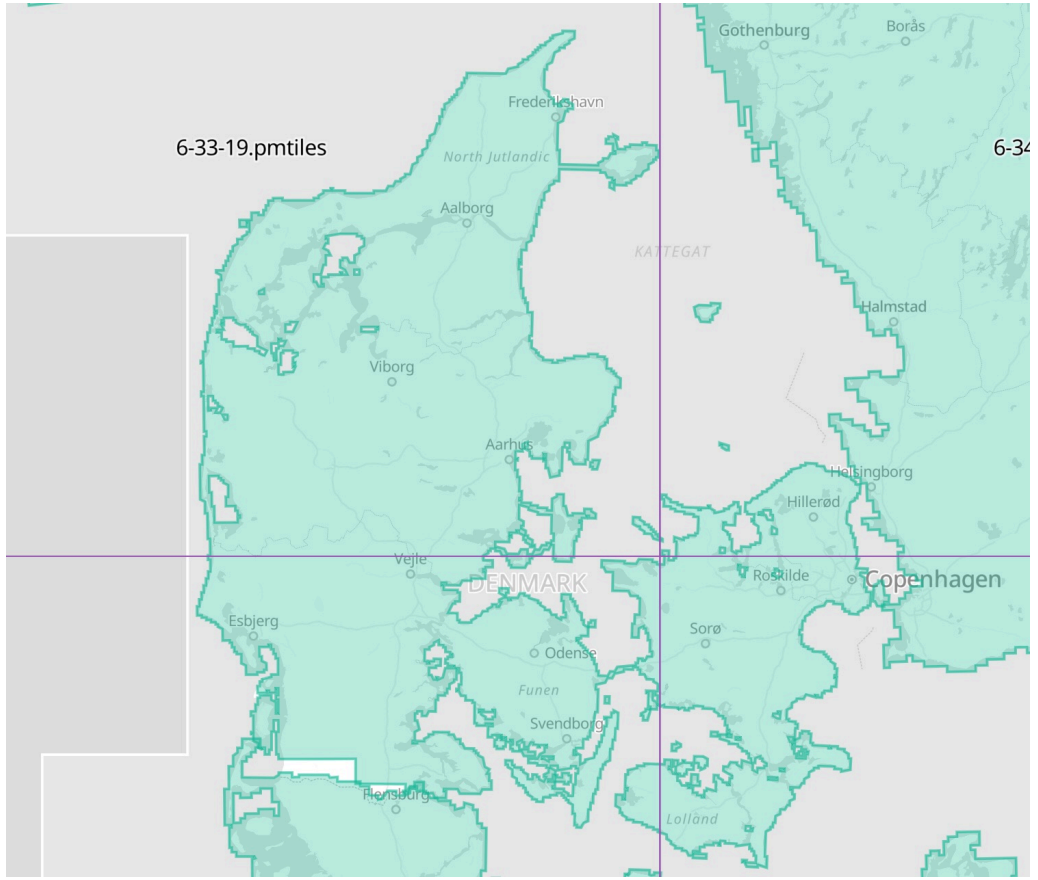
- Public STAC API
  - But rate limited
  - 1 request per IP
  - 70,000 tifs
  - 24 hours to read STAC
- Download password protected
  - Credentials via bad platform
  - Took 10 emails to setup



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Denmark

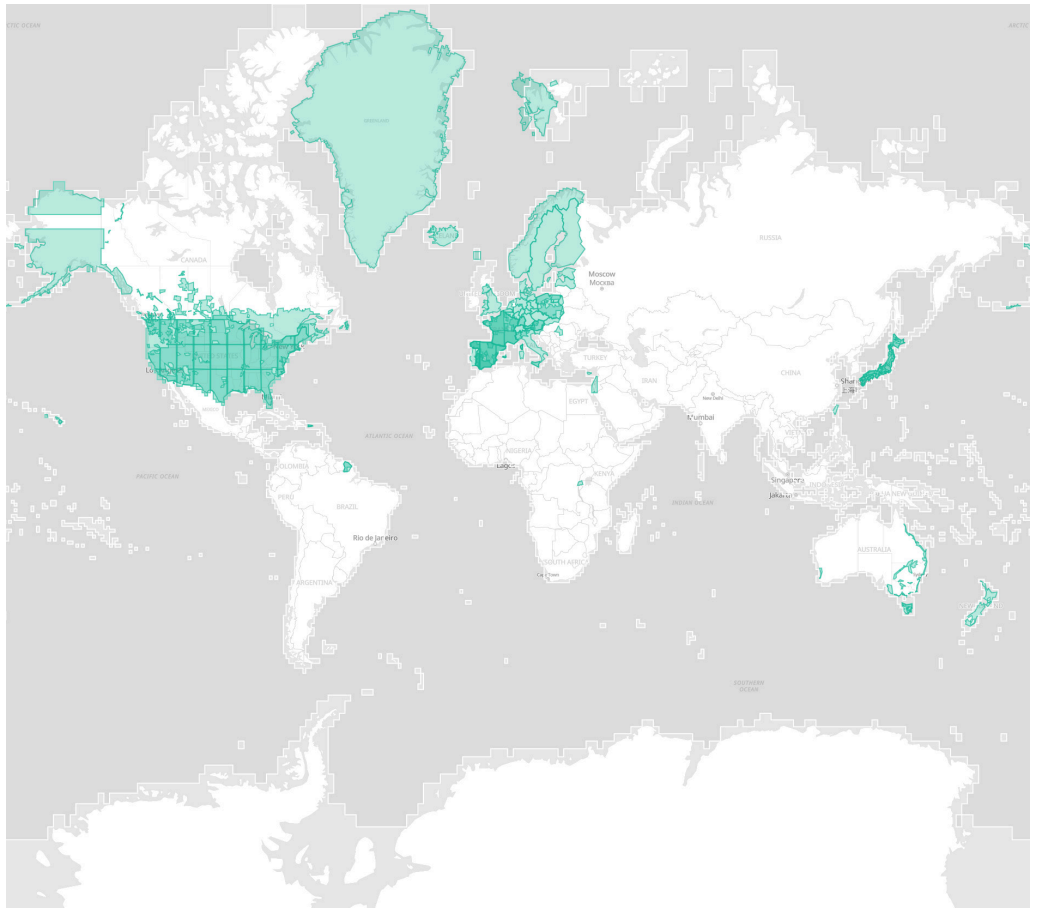
- Password protected FTP
- Credentials via good platform



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Global Coverage

- Global: Copernicus 30 m resolution
- Local: High-resolution Lidar
- 70 data producers
- CC-BY "or less"



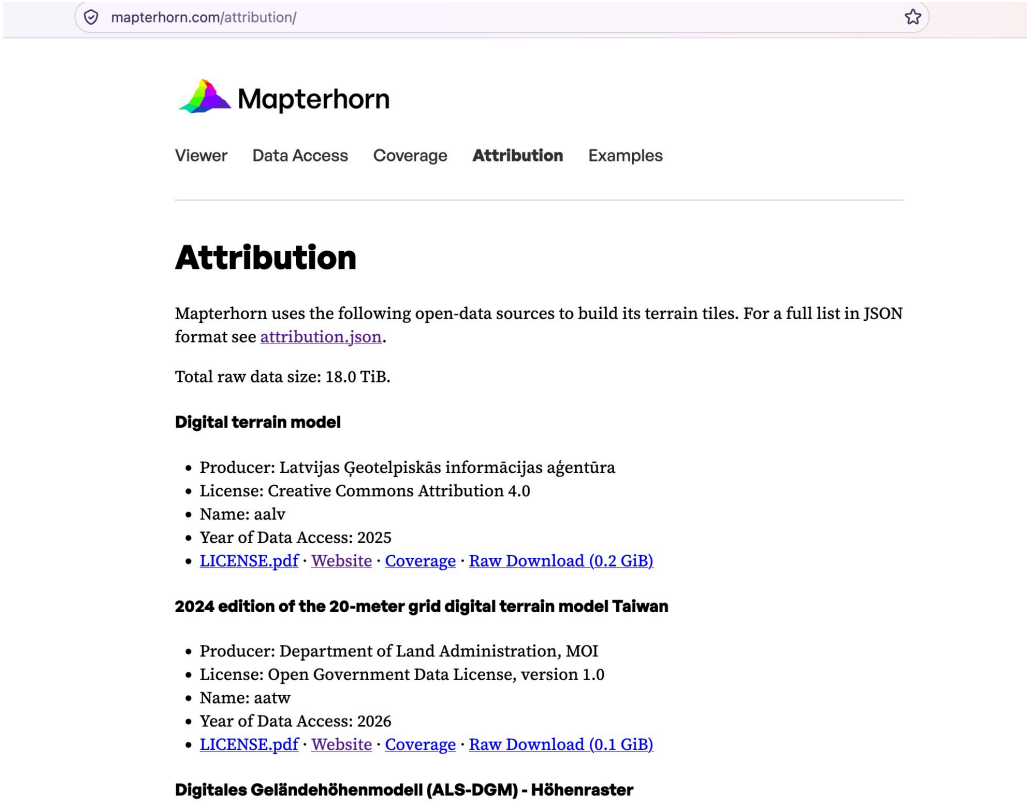
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Hierarchy of Terrain Data Formats

1. TXT
2. TXT in Zipfile
3. TIF uncompressed
4. TIF uncompressed in Zipfile
5. TIF compressed
6. TIF compressed with nodata, projection, internal tiling

Mapterhorn Output: Source tarballs

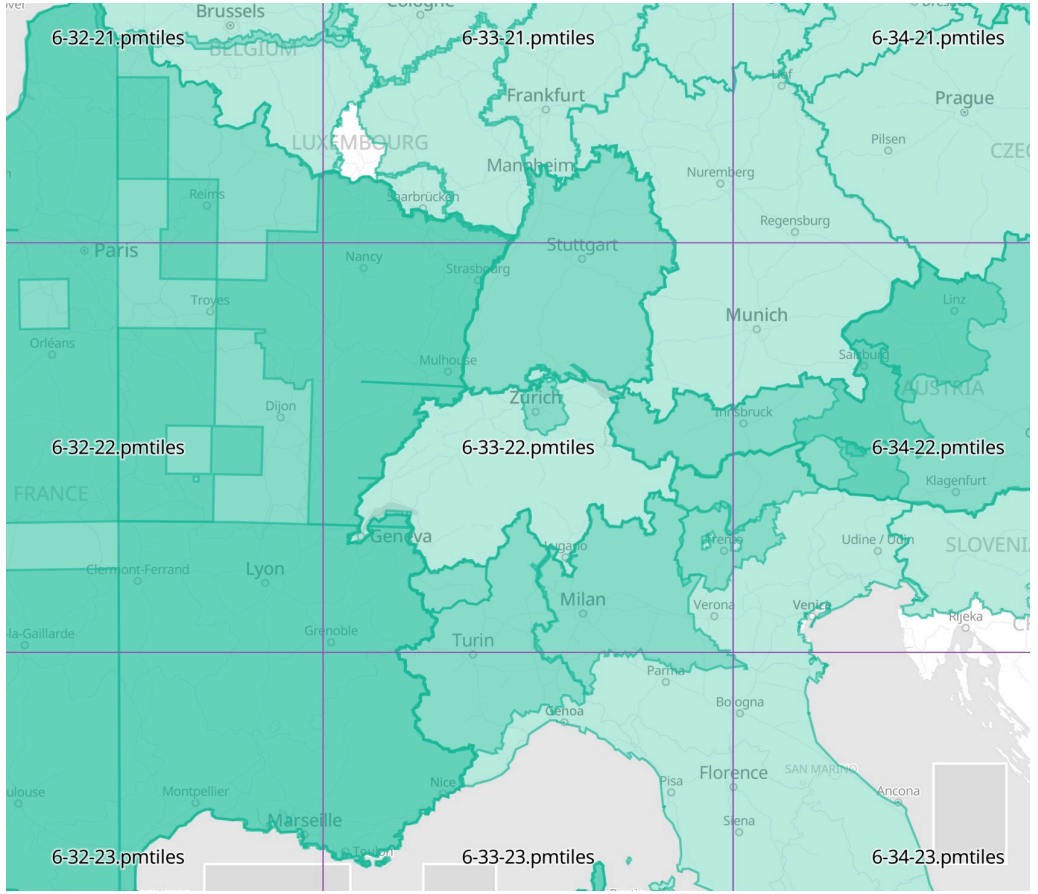
- Source tarballs with:
  - normalized TIFs
  - license
  - metadata
- 151 tarballs
- 18 TiB



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Mapterhorn Output: PMTiles

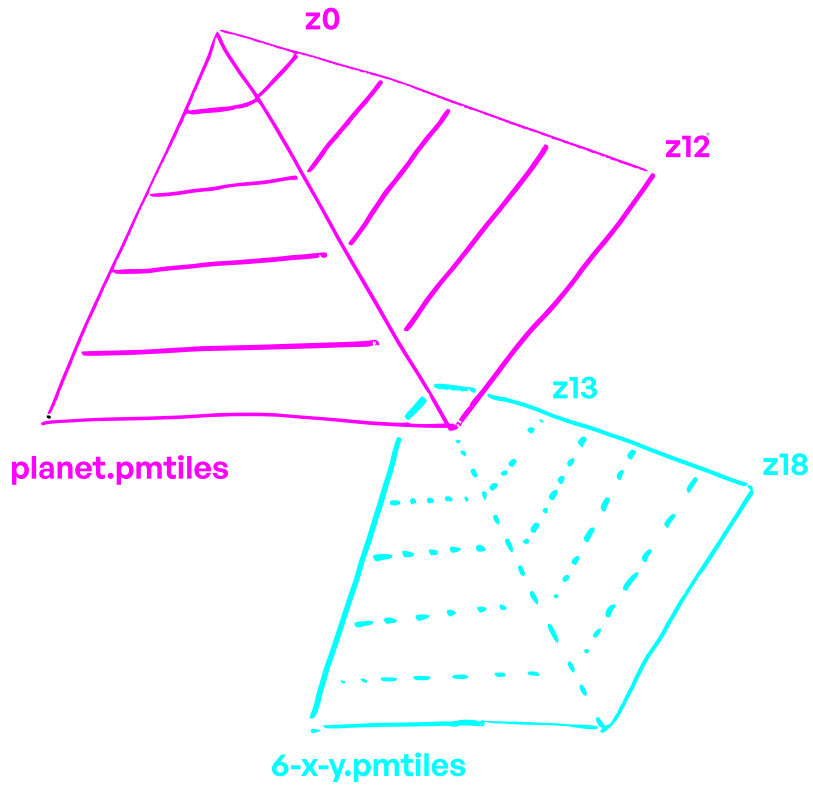
- Problem 1: S3 objects max 5 TiB
- Problem 2: variable resolution



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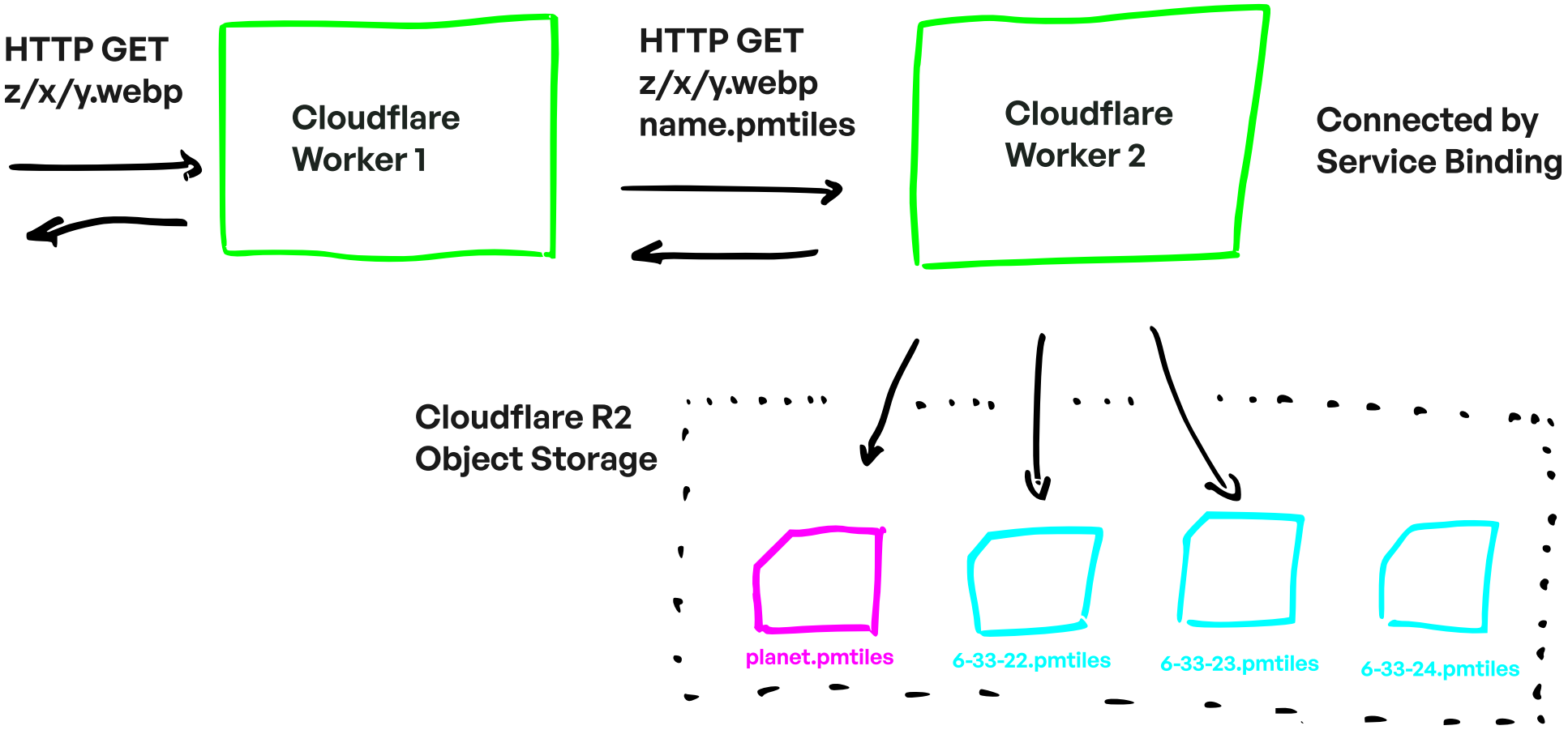
Mapterhorn Output: PMTiles

- planet.pmtiles:
  - global
  - zoom 0 to 12
- 6-x-y.pmtiles:
  - local
  - zoom 13+
- 11.8 TiB, 458 files



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tiles.mapterhorn.com



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Governance



Brandon Liu



Sascha Brawer



Oliver Wipfli

Full Access

- GitHub Organization
- Cloudflare Account
- Domain mapterhorn.com
- Financials at OpenCollective

Fiscal Host

OpenSourceCollective
501(c)(6) non-profit
440 N Barranca Ave #3939
Covina, CA 91723, USA

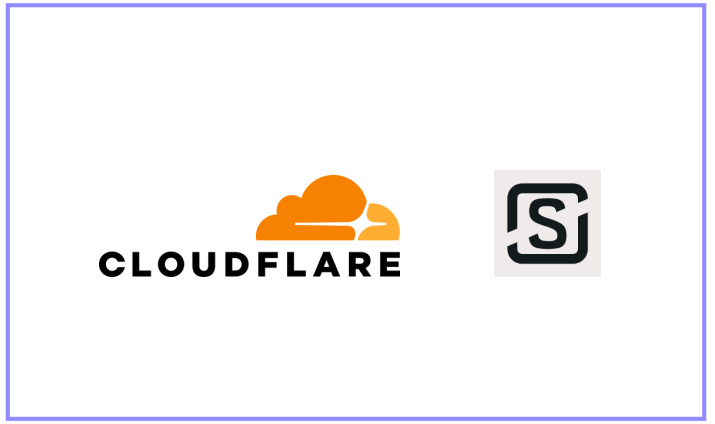
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Funding

opencollective.com/mapterhorn



Research & Development



Web Hosting



Maintenance

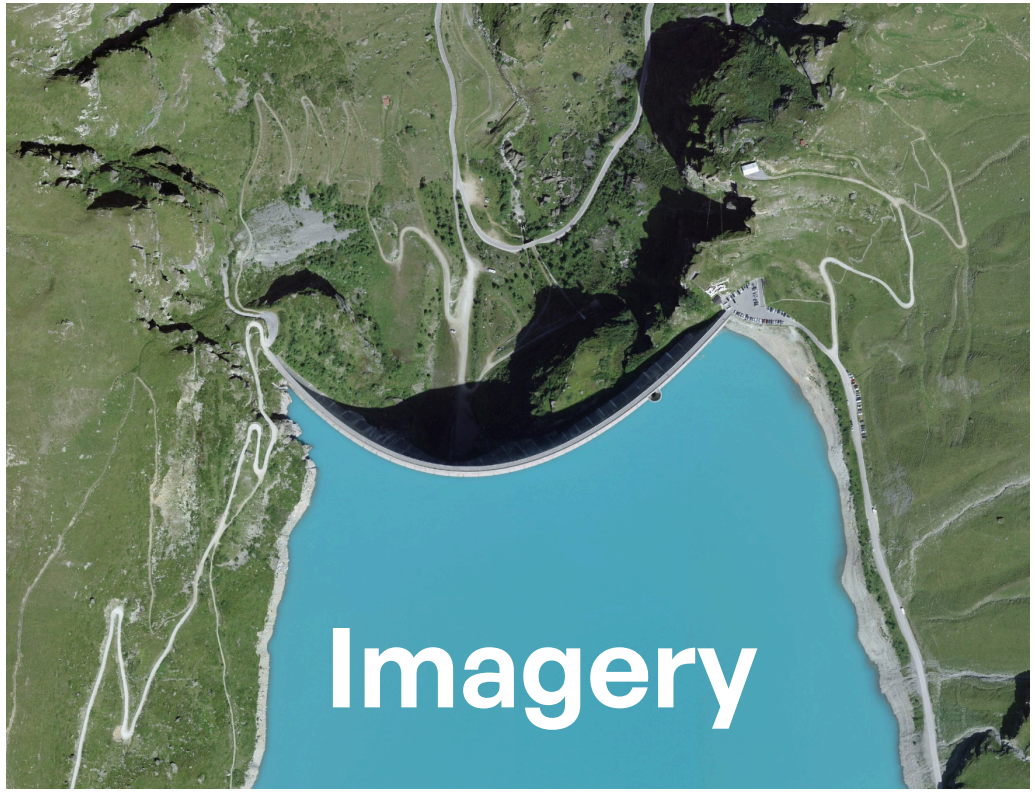
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Contributing

- Share bug reports, suggest new or updated datasets
  - https://github.com/mapterhorn/mapterhorn
- Contribute financially
  - https://opencollective.com/mapterhorn

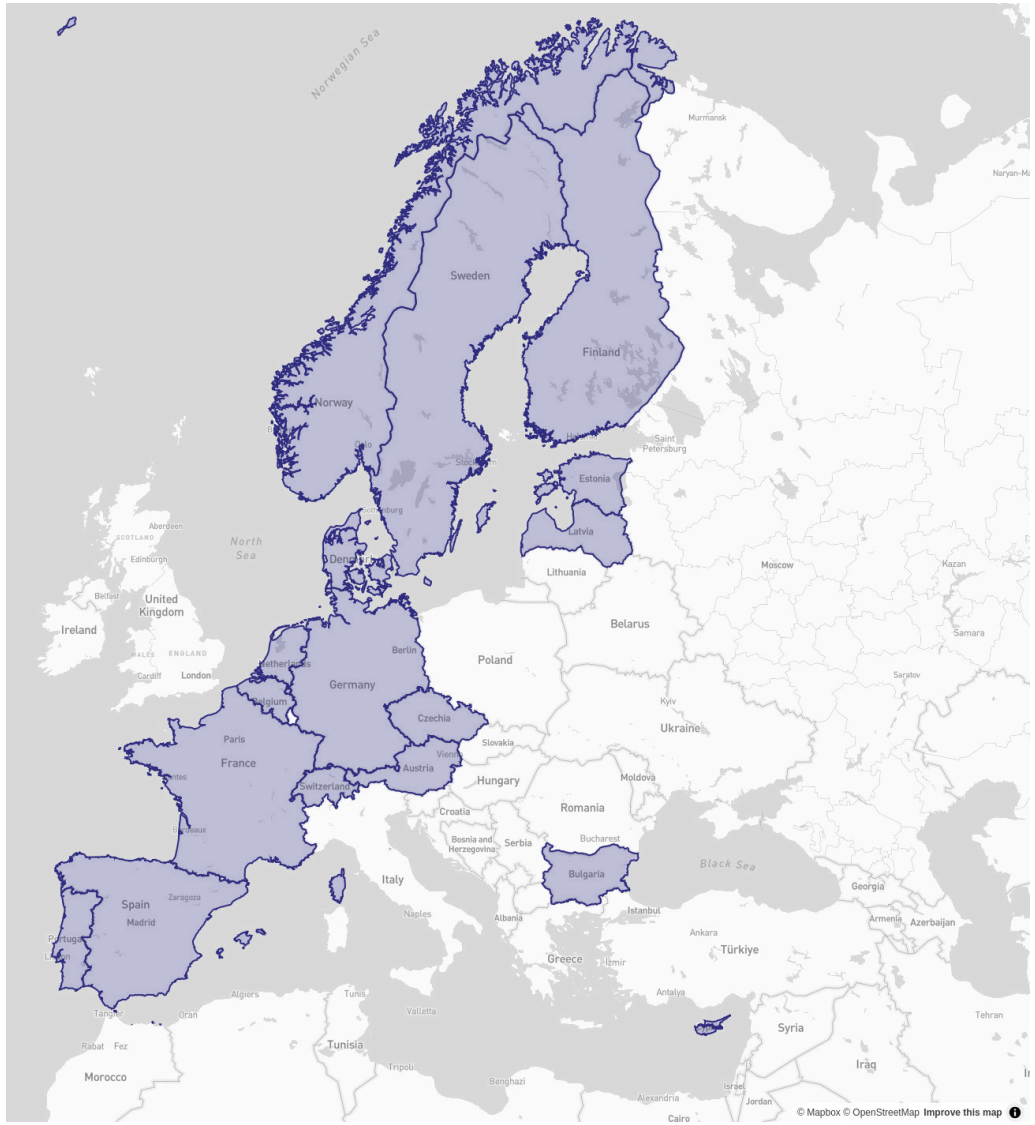
What's Next

- Aerial Imagery
- Same producers, "same" format, same pipeline
- Challenges:
  - 10x more data (100+ TiB)
  - Updates more frequently



Imagery Coverage

- Global: Sentinel-2 imagery 10 m
- High-resolution: 18 countries and regions
- Focus first on Europe, then expand to JP, NZ, US, CA, etc...
- First release still in 2026



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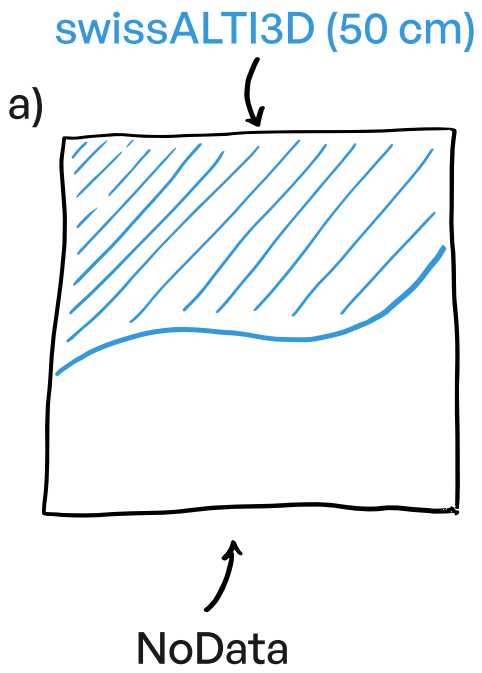
Thank You!

- Happy to chat: oliver.wipfli@leichteralsluft.ch



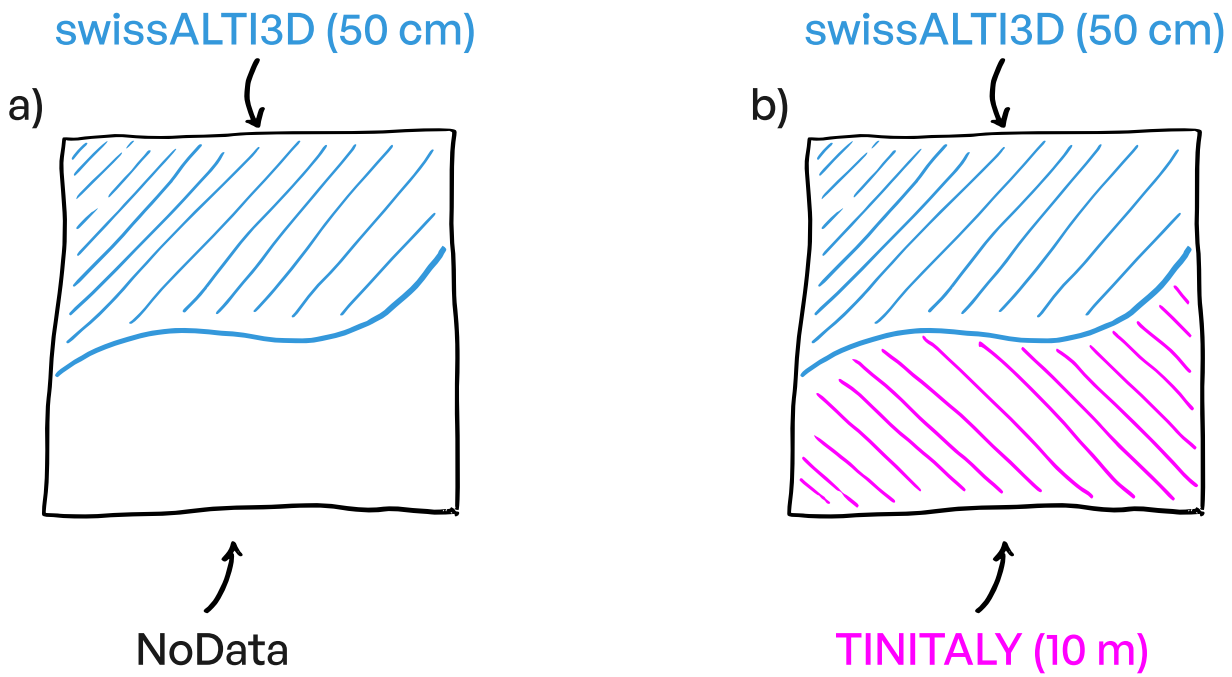
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Aggregation



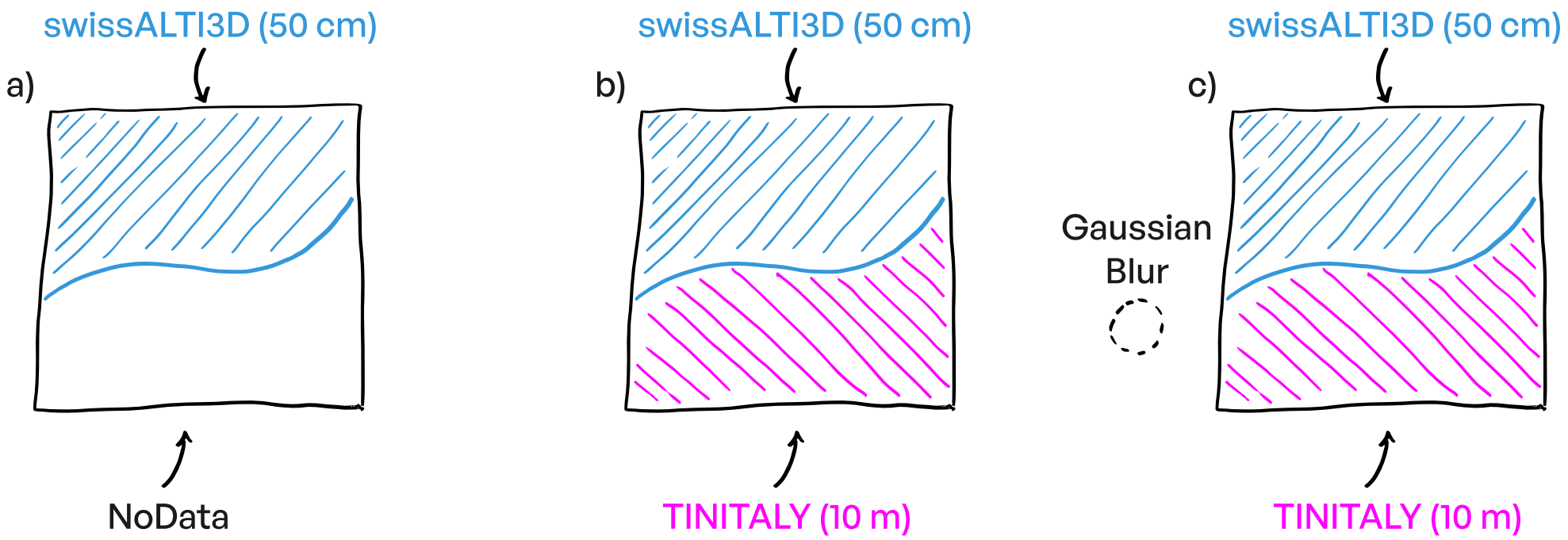
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Aggregation

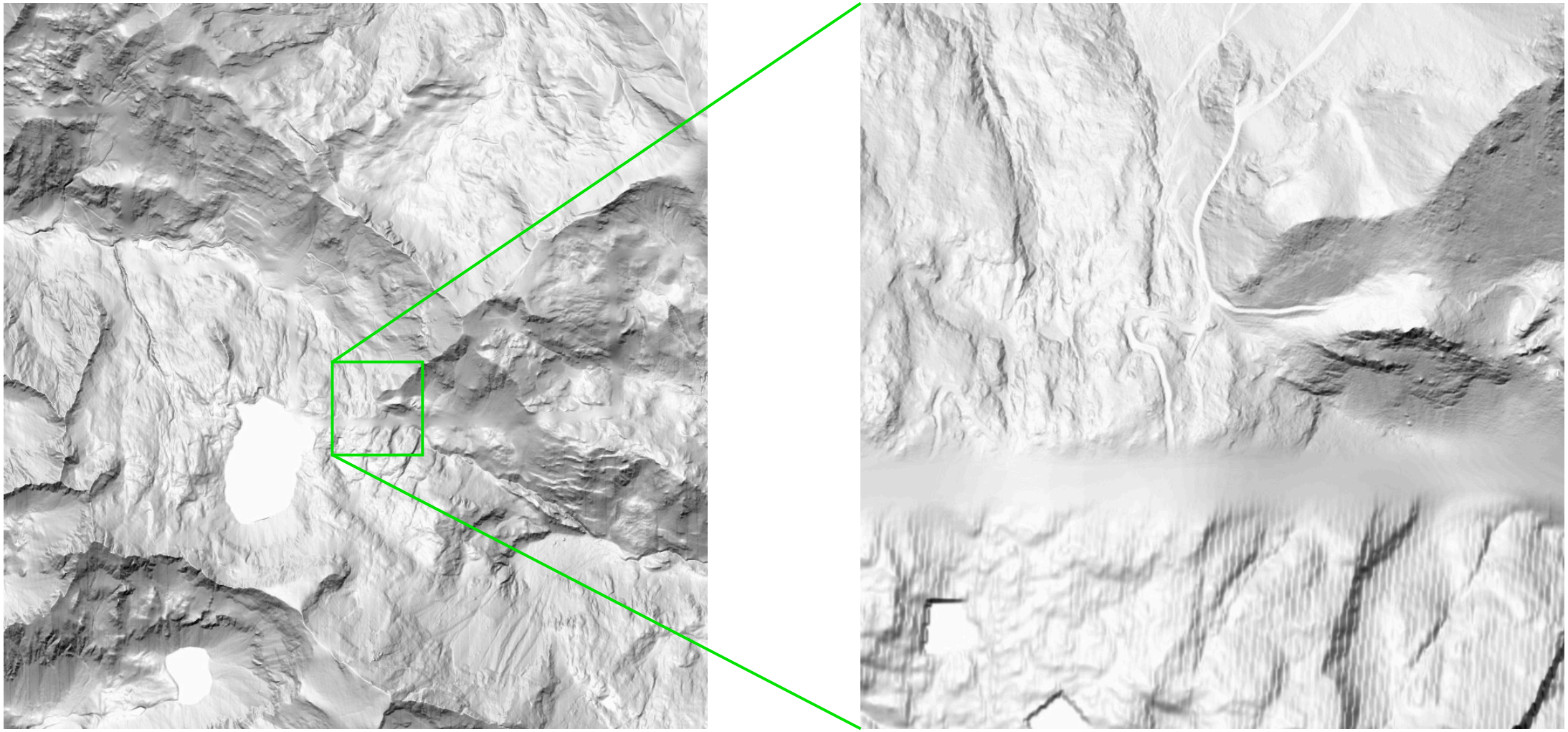


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Aggregation



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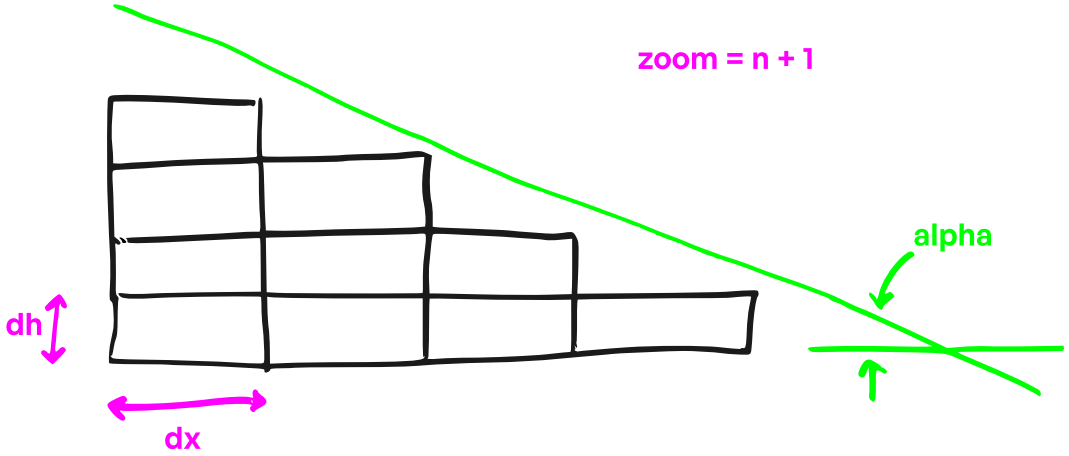
zoom 13.5

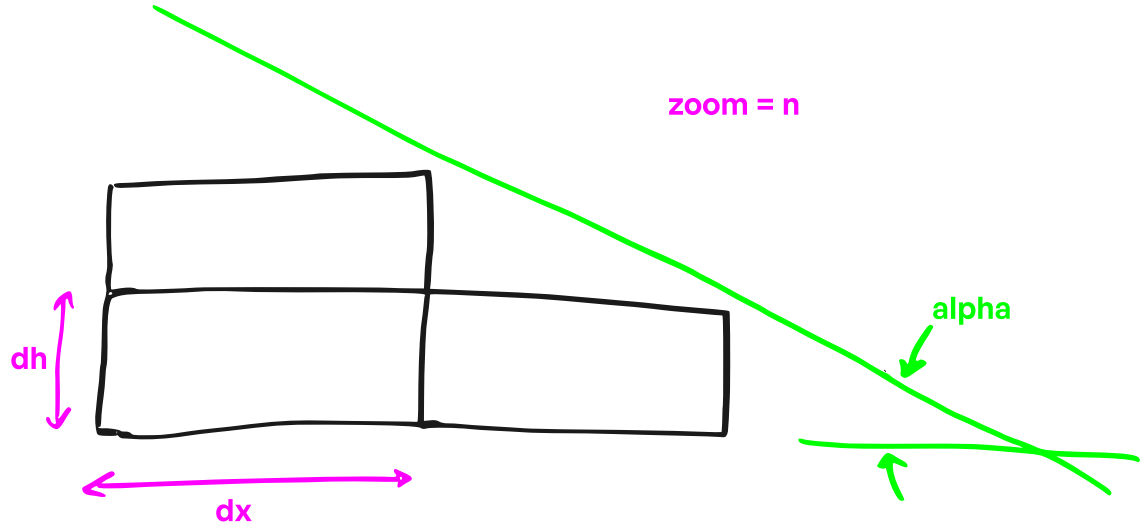
zoom 16.5

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Rounding

- Terrarium: elevation = 256 * red + green + blue / 256 - 2 ** 15
- zoom 19:
  - dx = 0.15
  - dh = 1 / 256
- zoom 18:
  - dx = 0.3
  - dh = 2 / 256
- zoom 17: ...

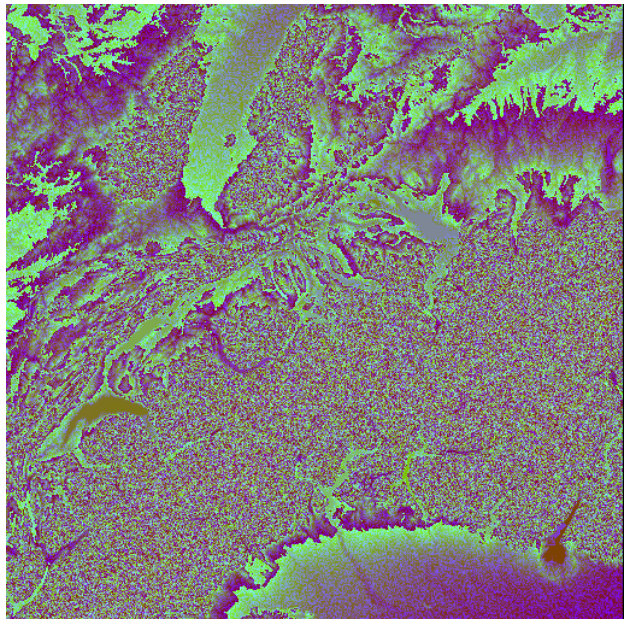




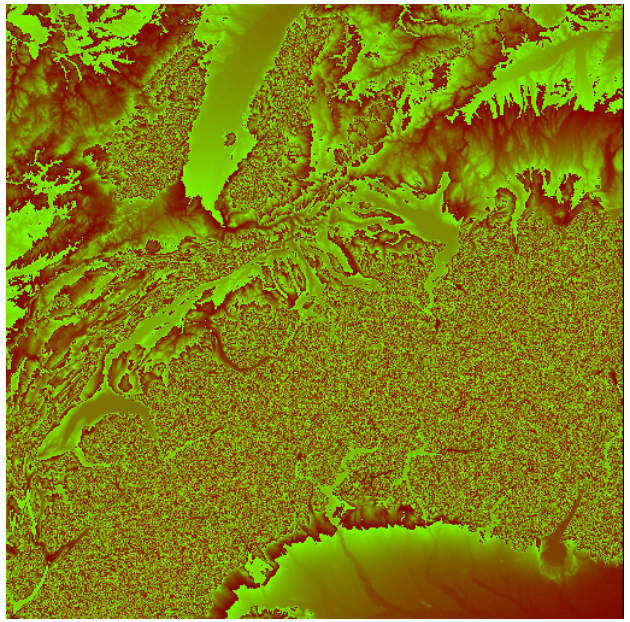
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Rounding

No Rounding
515 KiB









1 m Rounding
258 KiB

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Hardware Requirements

CPU

- Total number of tiles: 155M
- WebP lossless encoding ~200 ms per tile -> 359 days
- With 32 threads -> 11 days for WebP encoding alone, add 50% for reprojection and merging

RAM

- Only 200 MiB per thread -> 6.25 GiB for 32 threads
- Tune down GDAL_CACHEMAX and GDAL_MAX_DATASET_POOL_SIZE